CO2 ASSESSMENT

TOOL -01

APPLICATION BASED PROBLEM SOLVING

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a) Smart Online Examination Registration System - Block Box Testing.

b) The inputs are: Course Registration (yes/no), attendance (0-100%), fee payment (paid/unpaid) and a 12-digit numeric register number. Valid classes: registered, 75-100%, feepaid. Invalid classes: not registered, attendance below 75%, unpaid fee.

b) Equivalence class partitioning (ECP) tests one representative from each class. Boundary value analysis (BVA) focuses on the attendance threshhold: 74% → reject; 75% → pass attendance rule; 76% → pass.

c) Decision rules: (1) Course not registered → reject [2] attendance below 75 → reject (3) fee unpaid → request payment [4] invalid register number → display error [5] → All 4 conditions valid → accept.

d) Random can supplement planned tests. Example attendance values: 0, 23, 74, 75, 88, 101, -1. Register number examples: (123456789012) valid, (12345678901) (11digits not valid) and a 12-character value containing a letter. It could expose parsing errors and unanticipated failures.

e) Prioritize high-risk rules: the 75% threshold, invalid register number, unpaid fee. Use representative, boundary, negative and multi-failure data. Adequacy means all specified rules and boundaries are specified & exercised with the faults being recorded & no high severity defect remains unresolved.

22) Online Banking - State transition, Pairwise and core Testing.

a) State flow: Logged Out → (valid credentials + valid otp) logged in → (Valid beneficiary & amount) Transfer in Progress → success: Transfer Successful, Network failure: Transfer failed. From OTP verification, invalid attempts 1-2 remain in verification.

b) OTP Boundary Data: (1) Valid OTP on attempt 1 → login succeeds; (2) invalid OTP's, then valid OTP on attempt 2 → login succeeds. (3) Three invalid OTP's → account locks.

c) Pairwise Set: (covers all pairs across Device, Browser, Network):
1. Laptop - Chrome - WIFI
2. Laptop - Edge - Mobile Data
3. Mobile - Chrome - Mobile Data
4. Mobile - Edge - WiFi

d) Use-Case testing follows the transfer workflow: login, select beneficiary, enter amount, confirm process and verify receipt and balance. Exploratory testing interrupts the network, lets the session expire, double-click confirm, retries after timeout, and changes beneficiary details.

e) For the third-party core gateway, test API contracts, request/response mapping, success and decline codes, timeouts, retries, duplicate-request protection, compatibility across supported devices/browsers, availability and encrypted/authenticated communication.

a) Smart ATM Fund Transfer - CFG and Path Testing.

b) Control Flow Graph (CFG): Start/Read → D1: amount ≤ 0? Yes → "Invalid Amount" → Stop ; No → D2: amount > balance? Yes → "Insufficient Balance" → Stop ; No → D3: amount > daily limit? Yes → "Daily limit Exceeded" → Stop.

c) There are three decision nodes, so cyclomatic complexity V(G) = decisions + 1 = 3+1 = 4. Counting the compact CFG as 9 nodes & 11 edges, with one connected component: V(G) = E - N + 2P = 11 - 9 + 2(1) = 4. Therefore, atleast four independent paths are required for path testing.

d) Decision rules: [1] amount ≤ 0: amount = 0, balance = 1000, limit = 500 → invalid Amount. [2] amount > balance: amount = 800, balance = 500, limit = 1000 → Insufficient Balance. [3] amount > limit but ≤ balance: amount = 600, balance = 1000, limit = 500 → Daily limit Exceeded. [4] valid amount = 300, balance = 1000, limit = 500.

d) Statement coverage executes every statement atleast once. Branch coverage executes True and False out-comes of all three decisions. Condition coverage checks each Boolean condition as both True and false; the four tests above achieve this for the three simple conditions.

e) 100% statement coverage does not prove defect free software: a statement may run without its result being checked and missing requirements or data combinations may remain. Test adequacy also considers branches.